SUSPENSION

02
SECTION

02-00 OUTLINE

SUSPENSION ABBREVIATIONS

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4x2	4-Wheel 2-Drive
4x4	4-Wheel 4-Drive

SUSPENSION FEATURES

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Improved driveability	• Double wishbone suspension with torsion bar spring (front) has been adopted
Improved riding comfort	• Parallel arranged lower arm and tie rod (4x2 models) • Bias mount shock absorber (rear)
Improved serviceability	• Vehicle height adjustment by anchor bolt
Precision wheel alignment	• Camber and caster adjustment by shims

SUSPENSION SPECIFICATIONS[Australian Specs.]

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SPECIFICATIONS

Suspension

Item			4x2	4x4
Front suspension				
Suspension Type			Double wishbone	
Spring type			Torsion bar spring	
Stabilizer	Type		Torsion bar	
	Diameter	(mm {in})	Except Hi-Rider: 27 {1.06} Hi-Rider: 28 {0.94}	28 {0.94}
Shock absorber type			Cylindrical, double-acting	
Front wheel alignment (*Unloaded condition)	Maximum steering angle	Inner	Except Hi-Rider: 33°00′—37°00′ Hi-Rider:31°30′—35°30′	31°30′—35°30′
		Outer	Except Hi-Rider: 30°00′—35°00′ Hi-Rider:27°00′—32°00′	27°00′—32°00′
	Total toe-in	(mm {in})	2—8 {0.08—0.31}	3—9 {0.12—0.35}
	Camber angle		Regular cab: 0°35′ ±1° Hi-Rider (Stretch cab (with Rear Access System)): 0°44′ ±1° Hi-Rider (Double cab): 0°45′ ±1°	Regular cab and Stretch cab (with Rear Access System): 0°44′ ±1° Double cab: 0°45′ ±1°
	Caster angle		Regular cab: 1°56′±1° Hi-Rider (Stretch cab (with Rear Access System)): 2°07′±1° Hi-Rider (Double cab): 2°06′ ±1°	Regular cab and Stretch cab (with Rear Access System): 2°07′±1° Double cab: 2°06′±1°
	Steering axis inclination		Regular cab: 8°25′ Hi-Rider (Stretch cab (with Rear Access System)): 10°37′ Hi-Rider (Double cab)): 10°35′	Regular cab and Stretch cab (with Rear Access System): 10°37′ Double cab: 10°35′
Rear suspension				
Suspension Type			Leaf spring	
Spring type			Semi elliptic leaf spring	
Shock absorber type			Cylindrical, double-acting	

* : Fuel tank full; engine coolant and engine oil at specified level, and spare tire, jack, and tools in designated position.

Wheel and tires

Item			Specification			
Wheel	Size		15 × 61/2J		15 × 6 1/2J	16 × 7J
	Offset	(mm {in})	20 {0.79}		25 {0.98}	10 {0.39}
	Pitch circle diameter	(mm {in})	139.7 {5.500}			
	Material		Steel	Aluminum alloy	Steel	Aluminum alloy
Tire	Size		215/70R15C 106/104S		235/75R15 109S	245/70R16 111S

02-11 WHEEL ALIGNMENT

FRONT WHEEL ALIGNMENT DESCRIPTION

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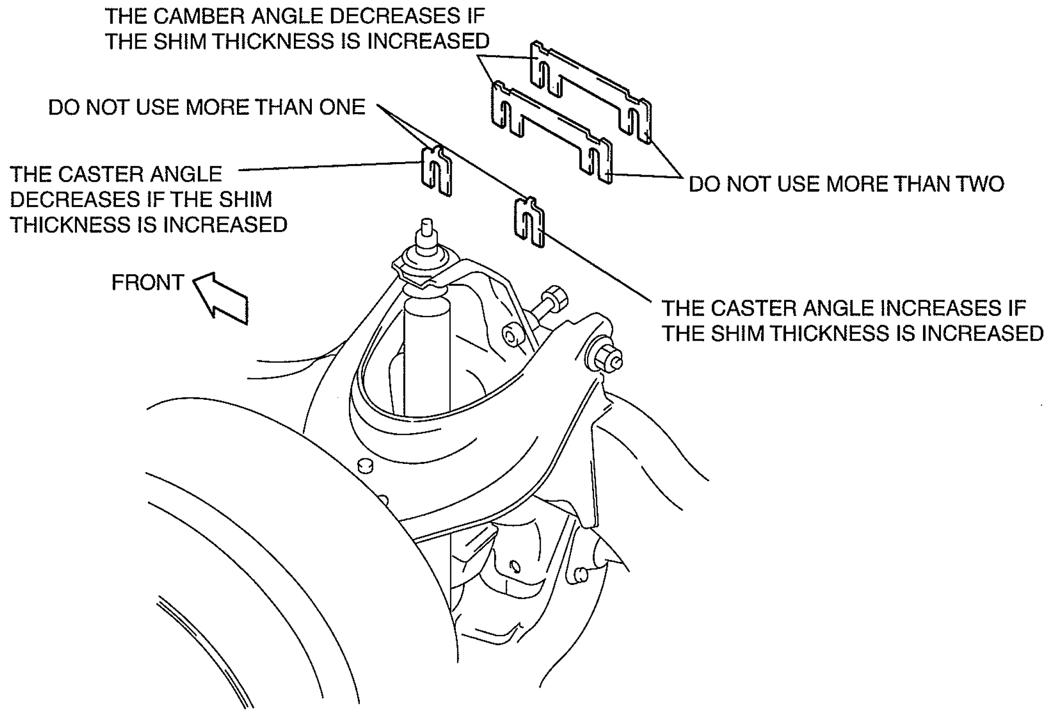
CAMBER AND CASTER



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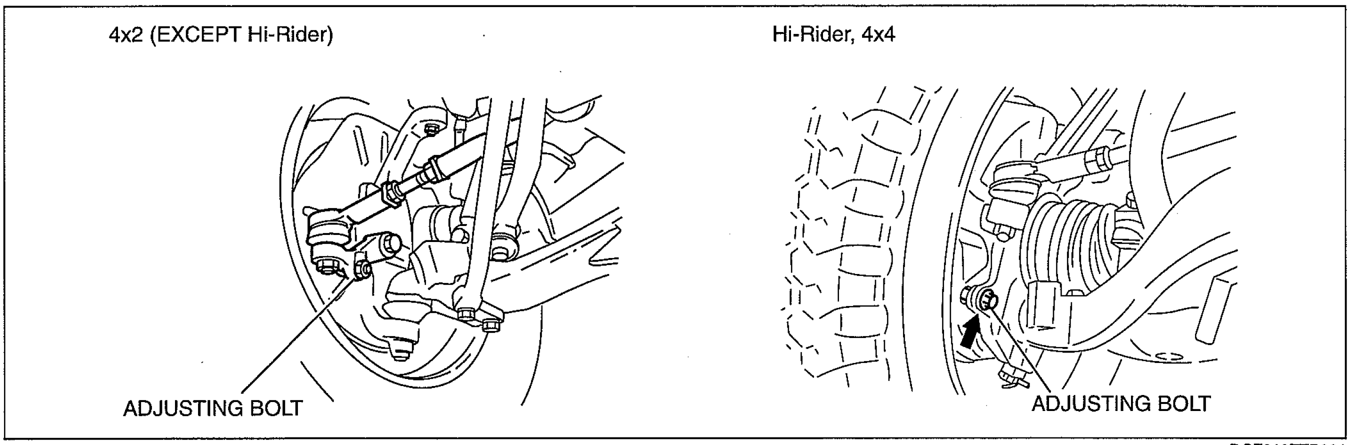
- To adjust the camber and caster angles, loosen the bolts of the upper arm shaft and insert or remove the adjustment shims.
- The amount of change of the camber and caster due to a **1 mm {0.039 in}** shim increase or decrease is as follows:

Camber: 1 mm {0.039 in} shim = 15'
Caster: 1 mm {0.039 in} shim = 30'

Shim

Thickness (mm {in})	
For camber adjustment	1.0 {0.039}
	1.6 {0.063}
	2.0 {0.079}
	3.2 {0.126}
	4.0 {0.157}
For caster adjustment	1.0 {0.039}
	1.6 {0.063}
	2.0 {0.079}
	3.2 {0.126}
	4.0 {0.157}
	0.6 {0.024}

MAXIMUM STEERING ANGLE



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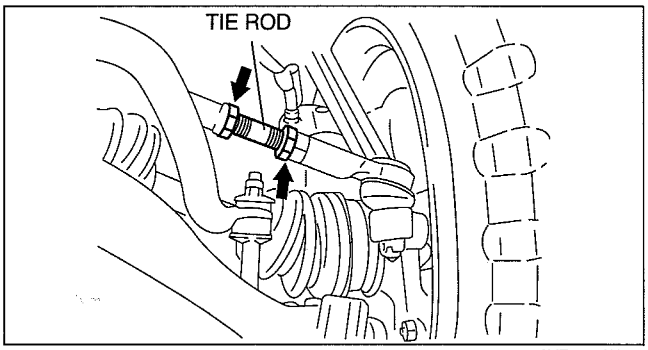
- The maximum steering angle can be adjusted by turning the adjusting bolts.

TOTAL TOE-IN

- The total toe-in can be adjusted by turning the tie rods.
- The left and right tie rods are both right threaded.To increase the toe-in, turn the right tie rod toward the front of the vehicle, and turn the left tie rod by the same amount toward the rear.
- One turn of the tie rod (both sides) changes the toe-in by **about 30 mm {1.18 in}**.



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02-12 WHEEL AND TIRES

WHEEL AND TIRES OUTLINE

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- In consideration of the environment, an adhesive-type balance weight made of steel has been adopted to reduce amount of lead used in the vehicle (aluminum alloy wheel).
- An adhesive-type balance weight is fastened on the inner side of the wheel. Since it is not visible from the styled side of the wheel, the design of the wheel is favoured.

WHEEL AND TIRES STRUCTURAL VIEW

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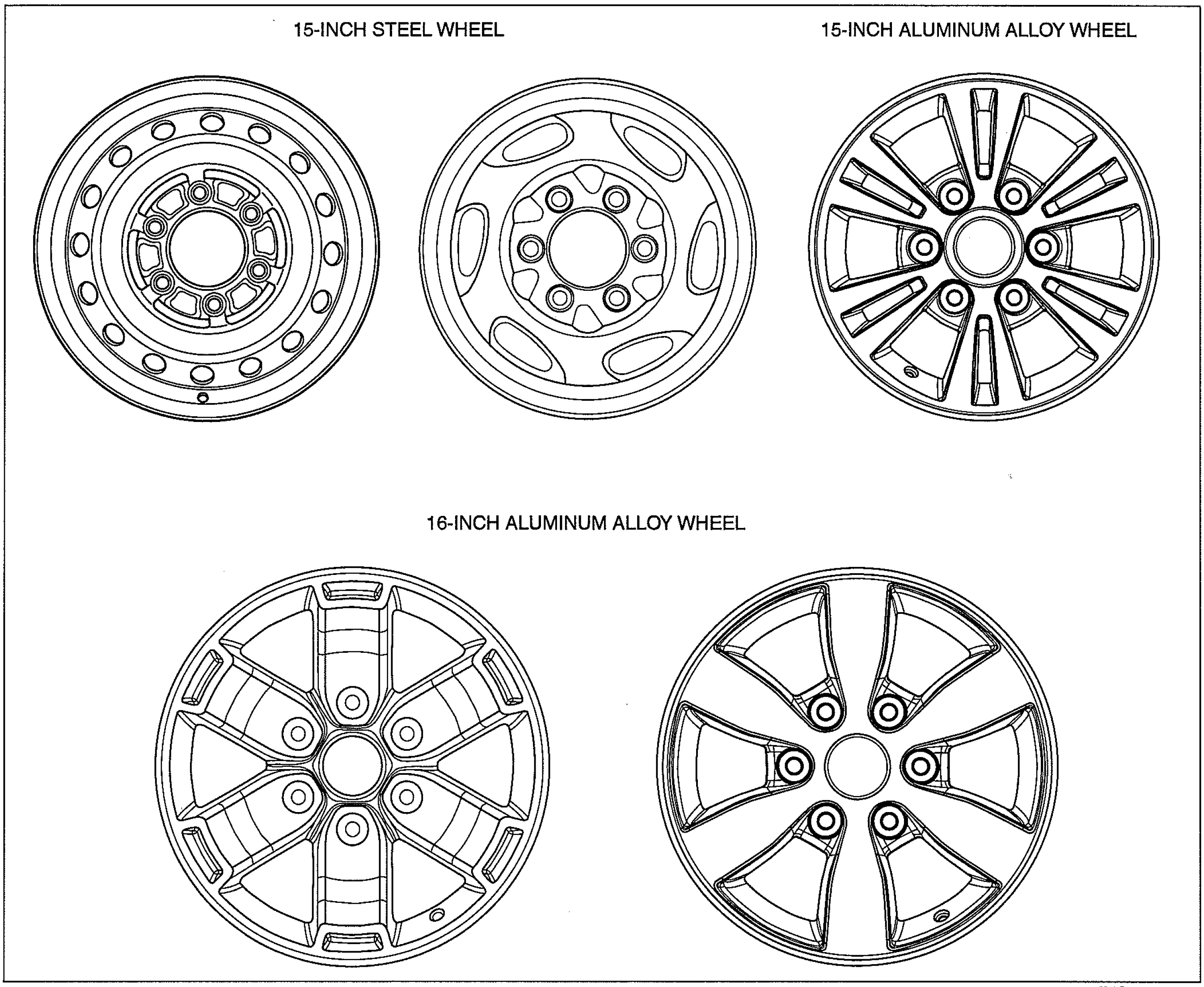

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02-13 FRONT SUSPENSION

FRONT SUSPENSION OUTLINE

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- The double wishbone suspension, which has I-shape lower arms and tension rods, is used for the front suspension of 4x2 models.
- The double wishbone suspension, which has A-shape lower arms, is used for the front suspension of Hi-Rider and 4x4 models. A-shape lower arms have high rigidly and durability.
- The torsion bar springs are used for both 4x2 and 4x4 models.
  The torsion bar spring is per-coiled in one direction. It therefore gives ample spring performance when installed correctly, but only half if installed in reverse.
  An R (right) or L (left) is marked on the rear end of the bar to distinguish one form the other.
  The left and right, and 4x2 and 4x4 torsion bar springs are not interchangeable.

FRONT SUSPENSION STRUCTURAL VIEW

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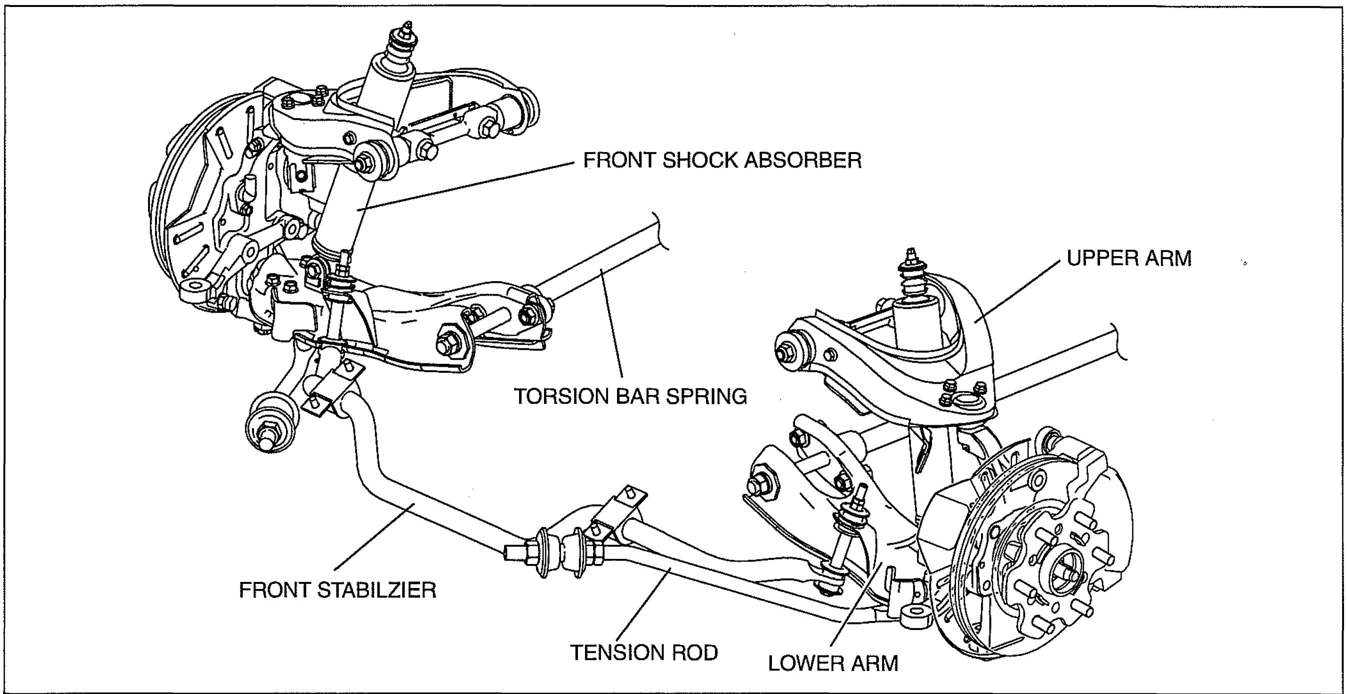
4x2 (Except Hi-Rider)



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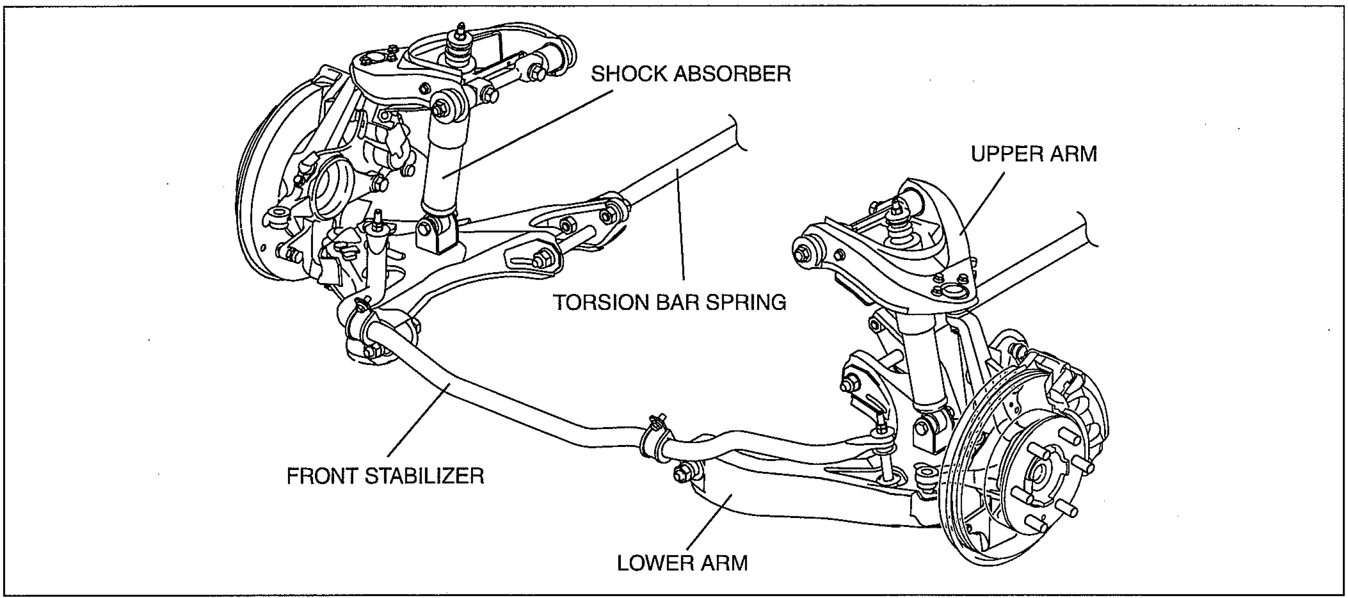
Hi-Rider



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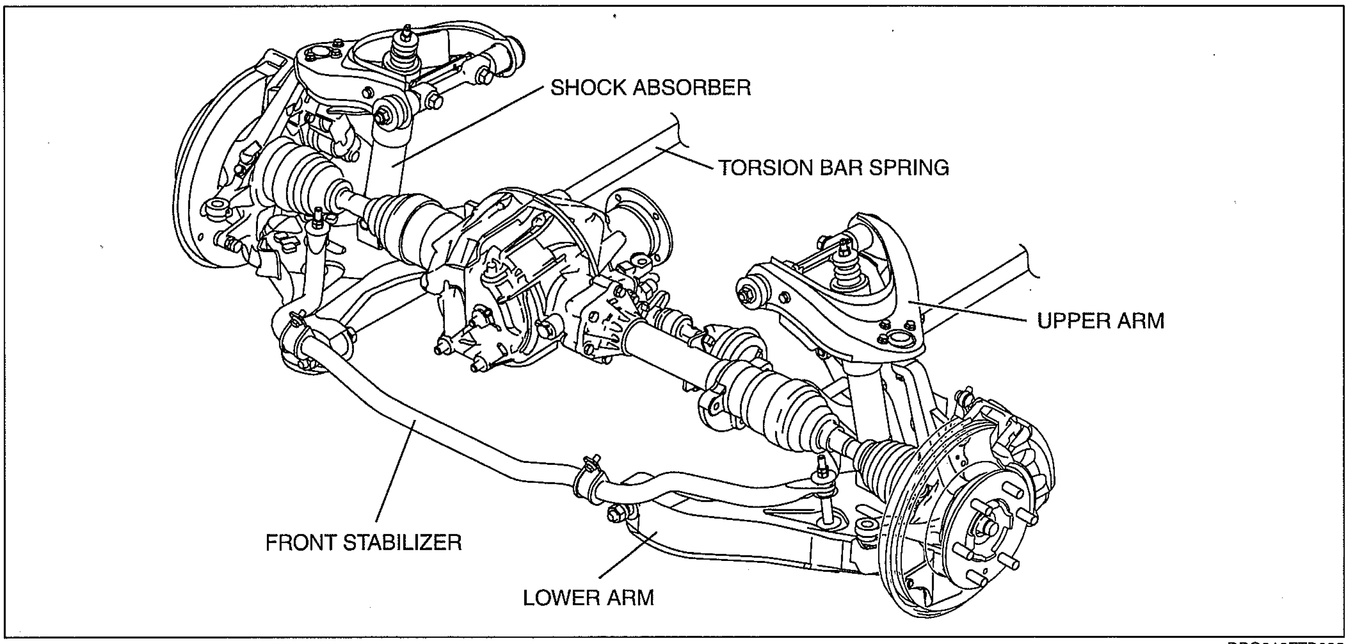
4x4



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FRONT SUSPENSION DESCRIPTION

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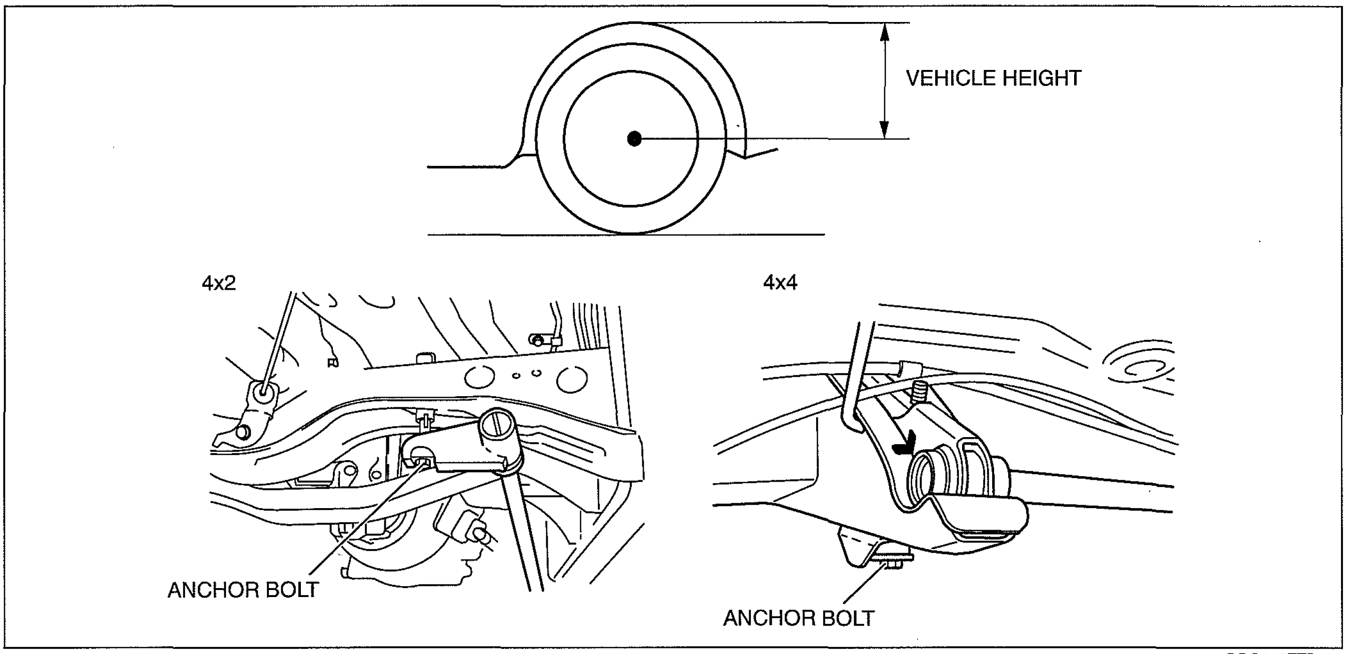
- The vehicle height can be adjusted by turning the torsion bar spring anchor bolt. When adjustment is required, adjust the vehicle height as follows:
  1. Place the vehicle on level ground.
  2. Inspect the front and rear tire pressure and adjust it if necessary.
  3. Measure the distance form the center of each front wheel to the fender brim.
     4x2 (Except Hi-Rider): **416—456 mm {16.4—17.9 in}**
     Hi-Rider: **512—552 mm {20.2—21.7 in}**
     4x4 (Stretch cab (with Rear Access Systetm)): **512—552 mm {20.2—21.7 in}**
     4x4 (Double cab): **502—542 mm {19.8—21.3 in}**
     *Difference between left and right: **10 mm {0.39in} max.**
  4. If the difference between left and right is not within the specification, adjust the vehicle height by turning the torsion bar spring anchor bolt.



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FRONT LOWER ARM AND TIE ROD CONSTRUCTION [4x2 (EXCEPT Hi-Rider)]

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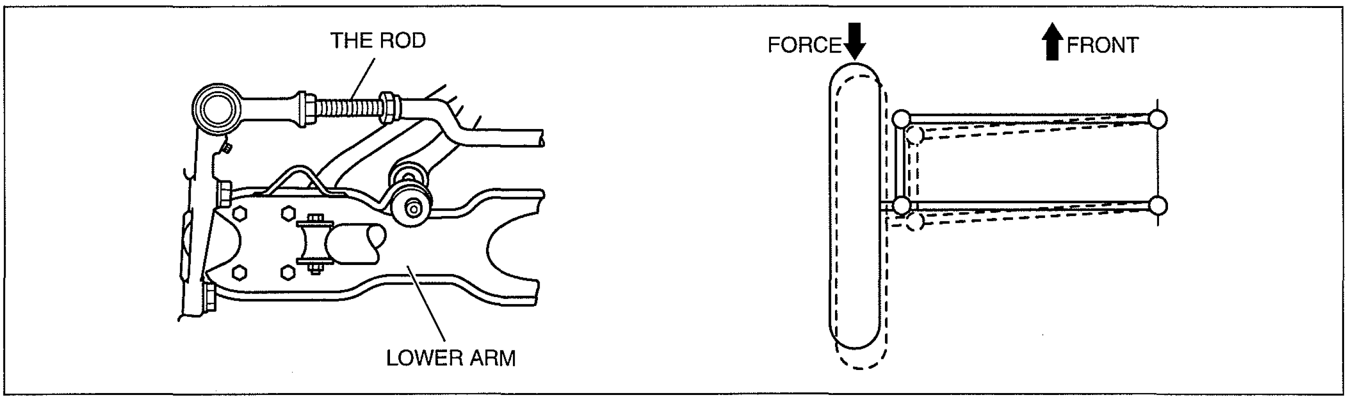
- To prevent shimmy, the lower arm and tie rod are arranged in parallel.



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- By aligning the lower arm and tie rod in parallel, a change in toe-in, which is the cause of shimmy, does not occur.
- The theory of why the tor-in change does not occur is that when force is applied to the tires form the front, they move backward, thus resulting in the rectangle (formed by the links) becoming a parallelogram configuration. As a result, the tires move in parallel to the inside and, although there is a slight change in the vehicle tread width, there is no change in tor-in.

02-14 REAR SUSPENSION

REAR SUSPENSION OUTLINE

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- The rigid axle suspension with leaf spring is used for the rear suspension.
- To prevent wind-up of the leaf springs during rapid acceleration or deceleration, the shock absorber on the right side is mounted at the rear of the axle and the left one is mounted at the front of the axle, (bias mount).In order to increase the rigidity in the lateral direction, wide shackle plates are used in the frame mounting.
- For Hi-Rider and 4x4 models, the rear springs are mounted on top of the axle housing to raise the center of gravity. Thus, the shock absorbers attach directly to brackets welded to the axle housing.

REAR SUSPENSION STRUCTURAL VIEW

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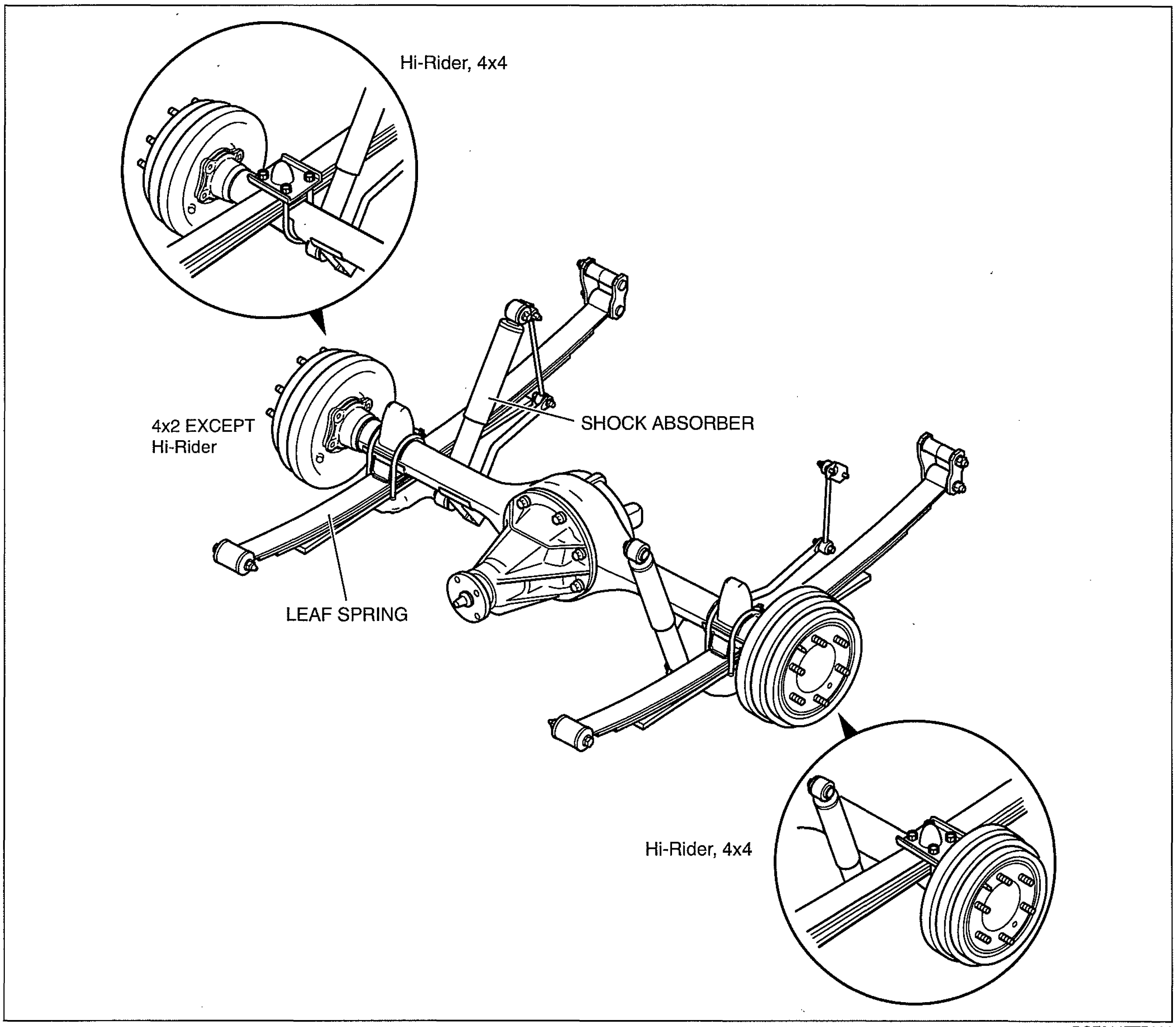

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